Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Getting Started
- Course Introduction — Video • 1 min
- Course Overview — Reading • 2 min
- Syllabus — Reading • 10 min
- Meet Your Instructor — Video • 1 min
- Meet Your Fellow Learners — Discussion Prompt • 10 min
- Academic Integrity — Reading • 1 min

Lesson 1: Introduction to Linear Algebra
- Introduction to Machine Learning — Reading • 25 min
- Introduction to Linear Algebra — Reading • 4 min
- Why Linear Algebra and Mathematics? — Reading • 2 min
- Notation — Reading • 5 min

Lesson 2: Systems of Linear Equations
- Foundational Concepts of Systems of Linear Equations — Reading • 20 min
- [HSP] System of Linear Equations — Ungraded App Item • 20 min
- Solved Example and Recommended Resources — Reading • 20 min
- Check Your Knowledge: System of Linear Equations — Graded Assignment • Grade: 100%

Lesson 3: Matrices
- Matrices and Matrix Operations — Reading • 20 min
- Matrices — Video • 8 min
- **Helpful Resources and Solved Examples** — Reading • 10 min
- Check Your Knowledge: Matrices — Graded Assignment • 16 min

Lesson 4: Vector Spaces
- Foundations of Vector Spaces: Operations and Subspaces — Reading • 20 min
- Vector Space — Video • 7 min
- Vector Space Properties — Reading • 10 min
- Subspace — Reading • 1 min
- Check Your Knowledge: Vector Spaces — Graded Assignment • 14 min

Lesson 5: Orthogonal Complement and Projection
- Introduction to Orthogonal Complement — Reading • 20 min
- Introduction to Orthogonal Projections — Reading • 40 min
- Importance of Projections — Reading • 4 min
- Projection Onto One-Dimensional Subspaces (Lines) — Reading • 15 min
- Projection Onto General Subspaces — Reading • 10 min
- Projections Case Study — Reading • 10 min

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Helpful Resources and Solved Examples

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This illustration provides a comprehensive overview of basic matrix operations, including addition, multiplication, scalar multiplication, transposition, and the calculation of determinants. It visually demonstrates the formulas and properties associated with these operations, such as the commutative and associative properties of addition, the distributive property of multiplication, and the non-commutative nature of matrix multiplication.

Addition $(\mathbf{A}+\mathbf{B})_{ij} = \mathbf{A}_{ij} + \mathbf{B}_{ij}$ $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} = \begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} & a_{13}+b_{13} \\ a_{21}+b_{21} & a_{22}+b_{22} & a_{23}+b_{23} \\ a_{31}+b_{31} & a_{32}+b_{32} & a_{33}+b_{33} \end{bmatrix}$ $\mathbf{A}+\mathbf{B} = \mathbf{B}+\mathbf{A}$

Multiplication $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} = \begin{bmatrix} a_{11}\cdot b_{11}+a_{12}\cdot b_{21}+a_{13}\cdot b_{31} & a_{11}\cdot b_{12}+a_{12}\cdot b_{22}+a_{13}\cdot b_{32} \\ a_{21}\cdot b_{11}+a_{22}\cdot b_{21}+a_{23}\cdot b_{31} & a_{21}\cdot b_{12}+a_{22}\cdot b_{22}+a_{23}\cdot b_{32} \end{bmatrix}$

$[\mathbf{AB}]_{ij} = \sum_{r=1}^{n} A_{ir}B_{rj} = A_{i1}\,B_{1j} + A_{i2}\,B_{2j} + \ldots + A_{in}B_{nj}$ $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$ $(\mathbf{A}+\mathbf{B})\mathbf{C} = \mathbf{AC}+\mathbf{BC}$ $\mathbf{C}(\mathbf{A}+\mathbf{B}) = \mathbf{CA}+\mathbf{CB}$ $\mathbf{AB} \neq \mathbf{BA}$

Scalar multiplication $(k\mathbf{A})_{ij} = k\cdot\mathbf{A}_{ij}$ $k\cdot\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} k\cdot a_{11} & k\cdot a_{12} & k\cdot a_{13} \\ k\cdot a_{21} & k\cdot a_{22} & k\cdot a_{23} \\ k\cdot a_{31} & k\cdot a_{32} & k\cdot a_{33} \end{bmatrix}$

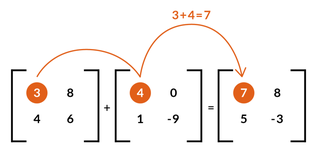
Transpose $(\mathbf{A}^{\mathrm{T}})_{ij} = \mathbf{A}_{ji}$ $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}^{\mathrm{T}} = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \\ a_{13} & a_{23} \end{bmatrix}$ **Square** $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}^2 = \begin{bmatrix} a_{11}^2+a_{12}\cdot a_{21} & a_{12}(a_{11}+a_{22}) \\ a_{21}(a_{11}+a_{22}) & a_{22}^2+a_{12}\cdot a_{21} \end{bmatrix}$

$(\mathbf{A}+\mathbf{B})^{\mathrm{T}} = \mathbf{A}^{\mathrm{T}}+\mathbf{B}^{\mathrm{T}}$ $(k\mathbf{A})^{\mathrm{T}} = k\left(\mathbf{A}^{\mathrm{T}}\right)$ $\left(\mathbf{A}^{\mathrm{T}}\right)^{\mathrm{T}} = \mathbf{A}$

Determinant $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}\cdot a_{22}\cdot a_{33} + a_{13}\cdot a_{32}\cdot a_{31} + a_{13}\cdot a_{21}\cdot a_{32} - a_{13}\cdot a_{22}\cdot a_{31} - a_{11}\cdot a_{23}\cdot a_{32} - a_{12}\cdot a_{21}\cdot a_{33}$

Solved Examples:

1. Addition



$$A = \begin{bmatrix} 1 & 3 & 5 \\ 7 & 9 & 11 \\ 13 & 15 & 17 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \\ 14 & 16 & 18 \end{bmatrix}$$

$$A+B = \begin{bmatrix} 1+2 & 3+4 & 5+6 \\ 7+8 & 9+10 & 11+12 \\ 13+14 & 15+16 & 17+18 \end{bmatrix} = \begin{bmatrix} 3 & 7 & 11 \\ 15 & 19 & 23 \\ 27 & 31 & 35 \end{bmatrix}$$

2. Subtraction

$$A = \begin{bmatrix} 6 & 5 & 4 \\ 3 & 2 & 1 \\ 0 & -1 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$A-B = \begin{bmatrix} 6-1 & 5-2 & 4-3 \\ 3-4 & 2-5 & 1-6 \\ 0-7 & -1-8 & -2-9 \end{bmatrix} = \begin{bmatrix} 5 & 3 & 1 \\ -1 & -3 & -5 \\ -7 & -9 & -11 \end{bmatrix}$$

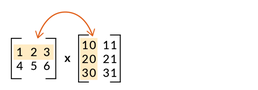
Note: You cannot add or subtract matrices when they do not have the same dimensions. For matrix addition or subtraction to be defined, both matrices must have the same number of rows and columns m × n.

To add or subtract matrices, they must have the same dimensions.

$$\underset{2\times 2}{\begin{bmatrix} 2 & 5 \\ -1 & -3 \end{bmatrix}} + \underset{3\times 2}{\begin{bmatrix} 4 & -2 \\ 0 & 7 \\ 0 & 0 \end{bmatrix}}$$

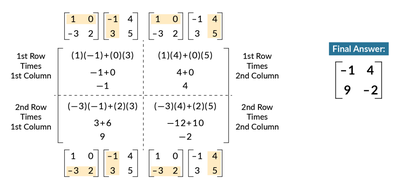
You cannot add them.

3. Multiplication



$$= \begin{bmatrix} 1x10+2x20+3x30 & 1x11+2x21+3x31 \\ 4x10+5x20+6x30 & 4x11+5x21+6x31 \end{bmatrix}$$

$$= \begin{bmatrix} 10+40+90 & 11+42+93 \\ 40+100+180 & 44+105+186 \end{bmatrix} = \begin{bmatrix} 140 & 146 \\ 320 & 335 \end{bmatrix}$$



4. Scalar Multiplication

$$k = 2, \quad A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$kA = 2\cdot\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 2\cdot 1 & 2\cdot 2 & 2\cdot 3 \\ 2\cdot 4 & 2\cdot 5 & 2\cdot 6 \\ 2\cdot 7 & 2\cdot 8 & 2\cdot 9 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \\ 14 & 16 & 18 \end{bmatrix}$$

Exercise

If C and D are given as:

$$C = \begin{bmatrix} m & -3 \\ -2 & -2n \end{bmatrix} \quad \text{and} \quad D = \begin{bmatrix} 6m & 3 \\ 2 & 4n \end{bmatrix}$$

and C+D=3I, where I is the identity matrix of order 2x2. Then find the values of m and n.

3I is the scalar multiple of the identity matrix I:

$$3I = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

It is given that C+D= 3 I:

$$\begin{bmatrix} m & -3 \\ -2 & -2n \end{bmatrix} + \begin{bmatrix} 6m & 3 \\ 2 & 4n \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} m+6m & -3+3 \\ -2+2 & -2n+4n \end{bmatrix} = \begin{bmatrix} 7m & 0 \\ 0 & 2n \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$$m = \frac{3}{7} \text{ and } n = \frac{3}{2}$$

5. Transpose:

If $M = \begin{bmatrix} 10 & -5 & -15 \\ 60 & 20 & -4\sqrt{2} \\ 8 & -9 & 45 \end{bmatrix}$ then the transpose M^T is: $M^T = \begin{bmatrix} 10 & 60 & 8 \\ -5 & 20 & -9 \\ -15 & -4\sqrt{2} & 45 \end{bmatrix}$

Now, if we take the transpose of $\left(M^T\right)^T$, we should get back the original matrix M:

$$\left(M^T\right)^T = \begin{bmatrix} 10 & -5 & -15 \\ 60 & 20 & -4\sqrt{2} \\ 8 & -9 & 45 \end{bmatrix} = M$$

6. Inverse:

If $C = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$

To find the inverse of this matrix, we need to check first that the inverse exists.

First, find the determinant of C:

$$|C| = \begin{vmatrix} 4 & 7 \\ 2 & 6 \end{vmatrix} = 4\cdot 6 - 7\cdot 2 = 24 - 14 = 10$$

Since |C|≠0, the inverse of C exists. Next, find the adjugate (adjoint) of C:

$$adj(C) = \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

Now, calculate the inverse of C:

$$C^{-1} = \frac{1}{|C|}adj(C) = \frac{1}{10}\begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} \frac{6}{10} & \frac{-7}{10} \\ \frac{-2}{10} & \frac{4}{10} \end{bmatrix}$$

$$C^{-1} = \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}$$

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